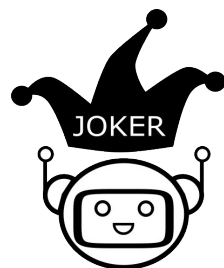


Searching for Sound–Meaning Collisions: Graph-Based Affordance Retrieval and Multi-Evaluator Ranking for Pun Translation

Russell Taylor · Adam Brikman · Prateek Awate

CLEF 2026, Jena, 23 September 2026

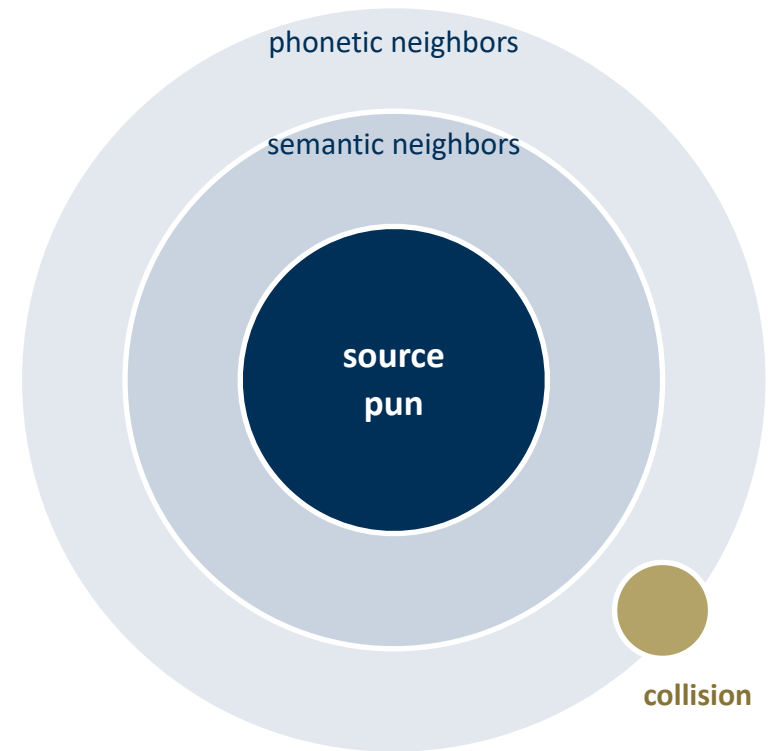


Pun translation is a search problem

Pun translation has been called impossible: the linguistic accident behind a pun rarely survives translation.

Low (2011): translators should stop searching for *equivalent words* and instead search for **new points of contact between sound and meaning**.

His squares, pentagons and hexagons describe a progressively expanding search through semantic and phonological neighborhoods. That is a procedure a computer can run.



From our 2025 system to 2026

2025

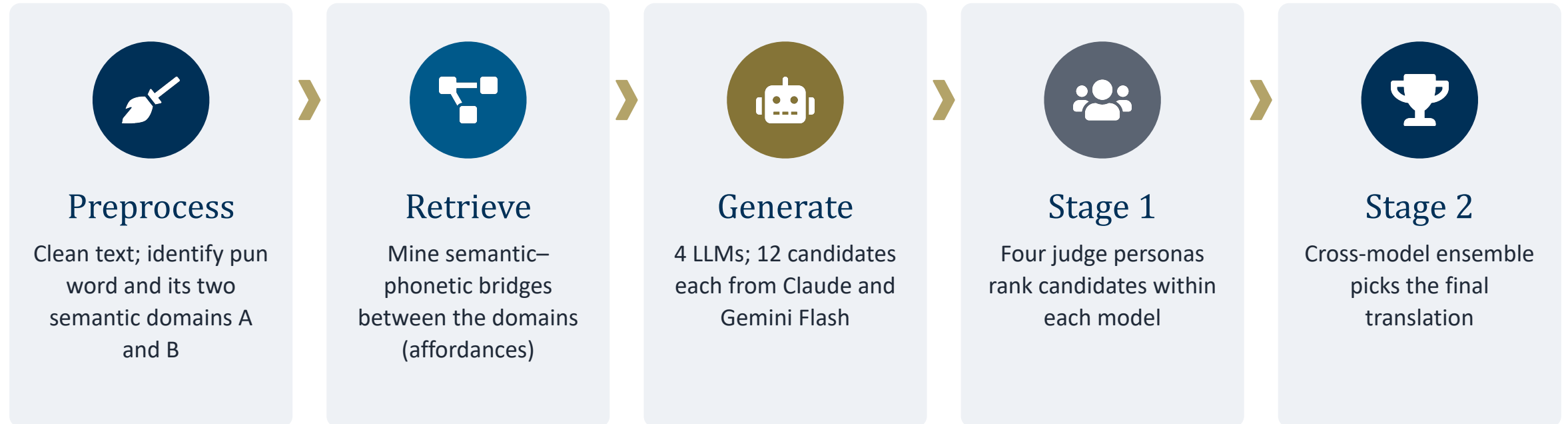
- 1st and 2nd in human evaluation (37 and 36 of 42)
- Evaluation-driven system beat retrieval-guided by one translation
- Word-level retrieval: phonetic + FastText, Low's first stage only
- Retrieval covered only 14% of puns
- One translation per input; low scores triggered regeneration

2026

- One pipeline: retrieval → generation → two-stage ranking
- Phrase-level expression bank + phonetic encoder trained on 2.7M relations
- Graph search through Low's later stages: 50.8% coverage
- 12 candidates per generator, ranked rather than regenerated
- Affordances traced through the whole pipeline

Open question: what role does retrieval actually play in successful pun translation?

Discover, explore, select



Retrieval uncovers hidden bridges · **generators** explore them · **evaluators** concentrate on the strongest collisions

Understand the source pun first



Data cleaning

- Normalize Unicode punctuation
- Repair malformed quotation and dialogue
- Keep hashtag cues that mark pun words
- Strip URLs, handles, emojis, reaction text
- ~100 dataset-specific repairs found by automated validation



Pun analysis

gemini-3-flash with structured JSON output returns:

Pun word

the word carrying the double meaning

Pun type

homophonic or homographic

Domain A

French word list for sense 1

Domain B

French word list for sense 2

From words to phrases: a French expression bank

370,450

French expressions indexed for dense retrieval

96.1%

are multi-word units: idioms, collocations and longer contexts that single words miss

Embedded with BAAI/bge-m3 (normalized), indexed with FAISS inner-product search

51,589

Wiktionary

entries labeled expressions, proverbs or locutions

12,308

PARSEME

sentence contexts with annotated multiword expressions

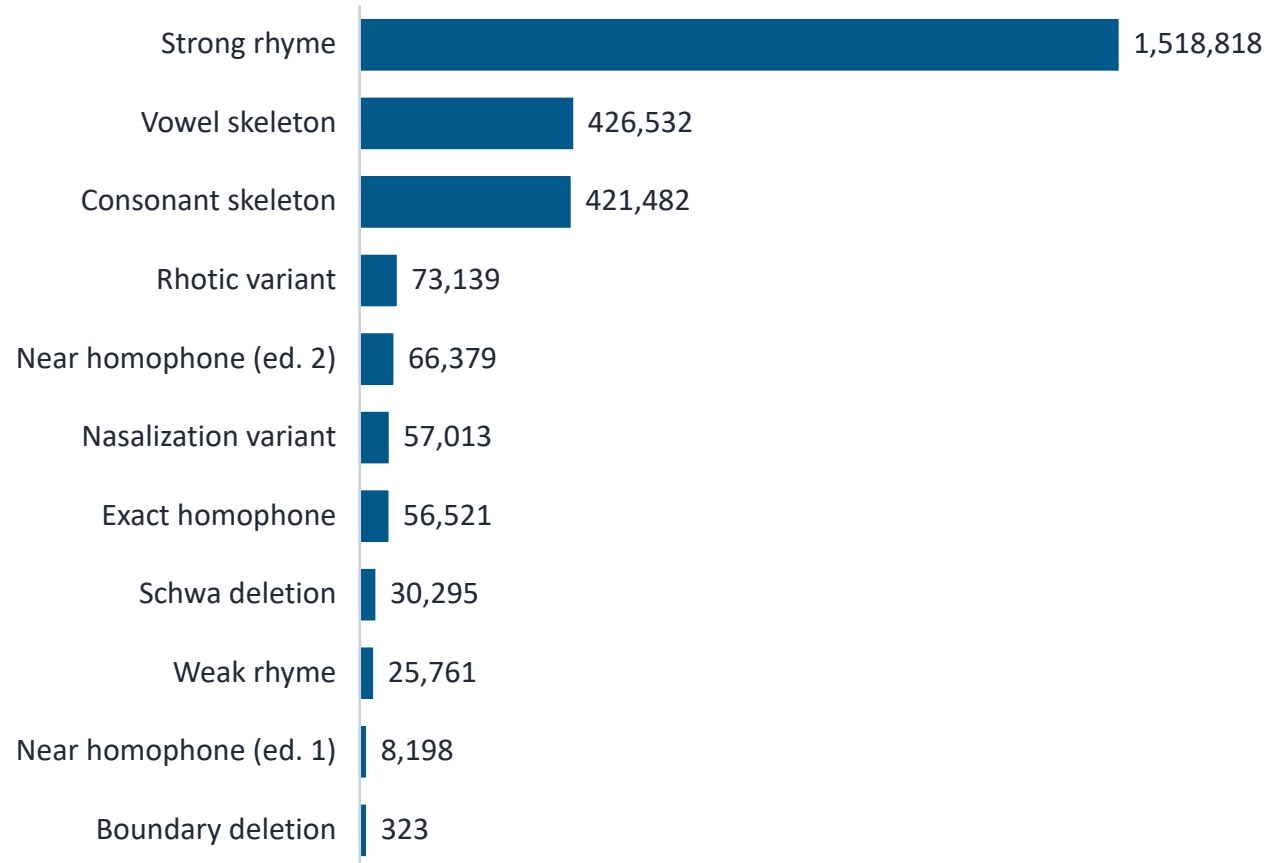
306,553

OpenSubtitles

high-association collocations (spaCy lemmas, POS patterns, frequency, PMI)

Teaching an encoder what “sounds close enough”

Positive relation types (2.68M pairs)



245,746

word–IPA pairs in the phonetic relevance graph

4.46M

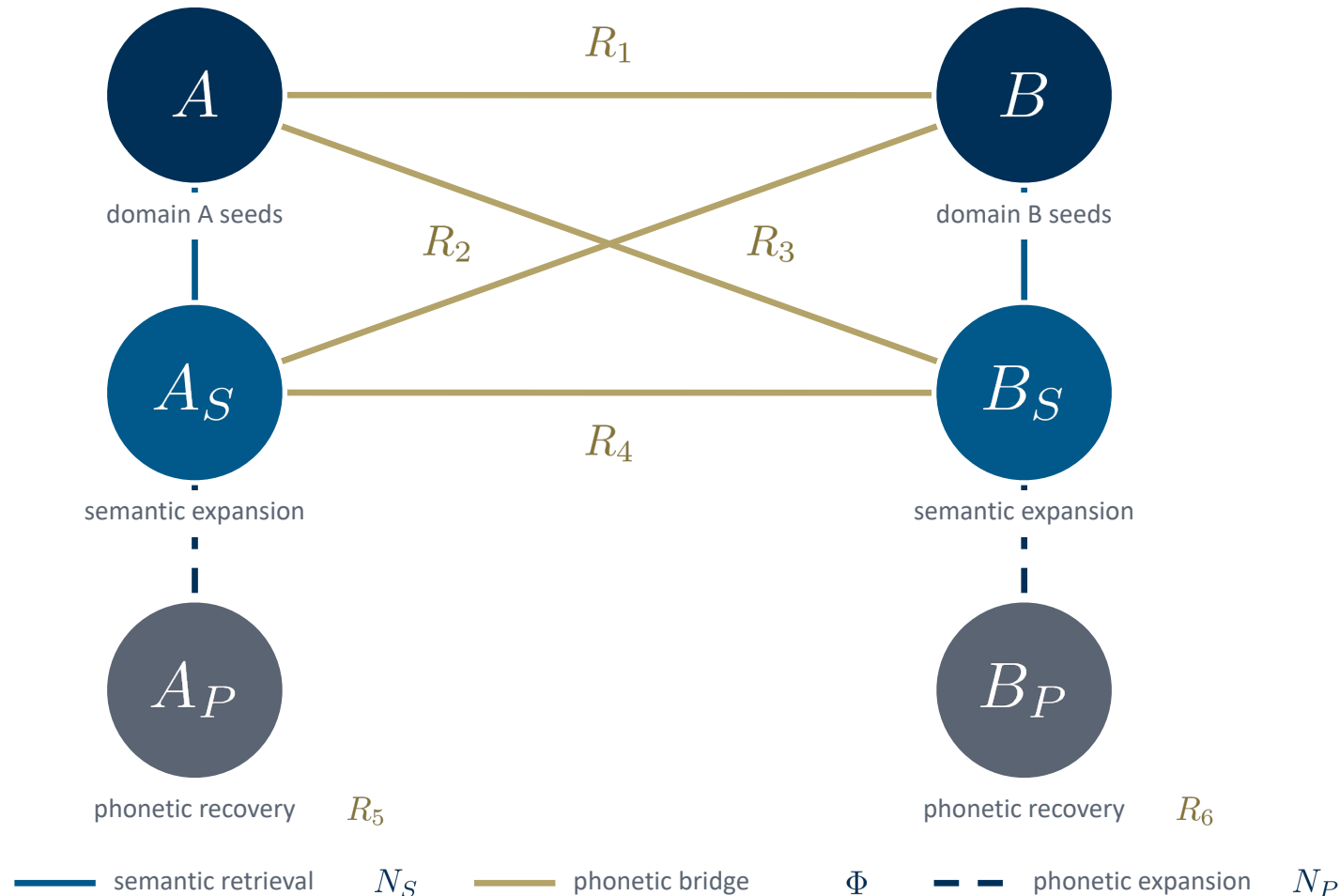
labeled phonological relations, with mined hard negatives

Fine-tuned bge-m3 on IPA with Multiple Negatives Ranking Loss; capped per-relation sampling keeps rhymes from dominating.

Recall@10 (6,321 queries)

Near homophone	0.642
Consonant skeleton	0.371
Hard negative	0.000

Mining bridges between two semantic regions



$$\Phi(X, Y) = \{(x, y) \in X \times Y : \cos(\mathbf{e}_x, \mathbf{e}_y) \geq \tau\}$$

$\mathbf{e}$ = learned phonetic (IPA) embedding

$$R_1 = \Phi(A, B)$$

$$R_2 = \Phi(A_S, B)$$

$$R_3 = \Phi(A, B_S)$$

$$R_4 = \Phi(A_S, B_S)$$

$$R_5 = \Phi(A_P, A \cup A_S)$$

$$R_6 = \Phi(B_P, B \cup B_S)$$

Ordered stages; a duplicate keeps the earliest stage that found it.

Keep the strongest, most diverse collisions



Phonetic

How similar is the pronunciation?



Semantic

How well does it anchor to the source domains?



Naturalness

Is it recognizable French lexical material?



Pivot

How useful is it for building a pun?

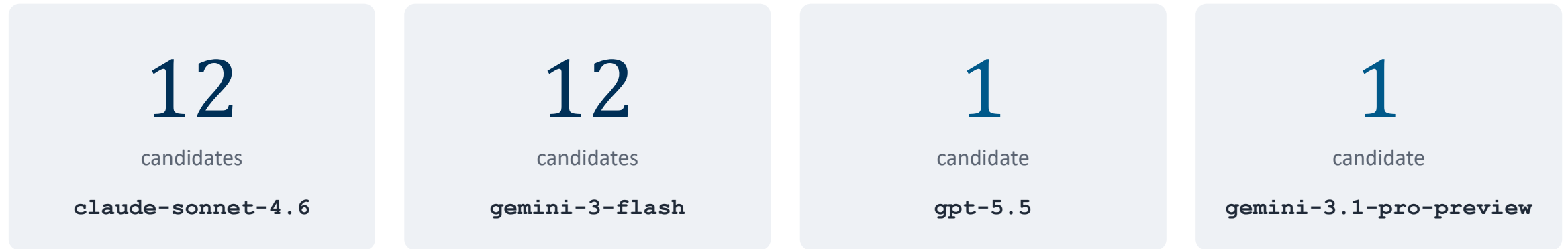
Pruning

- Drop exact duplicates
- Penalize inflectional, accent, plural and same-root echoes
- Per-surface and per-root caps stop one lexical family dominating
- Limit identity and other low-creativity collisions

14% → 50.8%

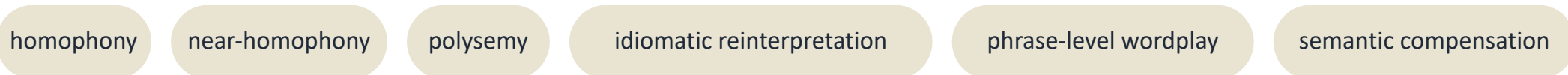
retrieval coverage, 1–6 affordances per covered pun

From one answer to many competing candidates



Input: English pun + translated semantic domains + retrieved affordances

Prompt encourages different forms of French wordplay:



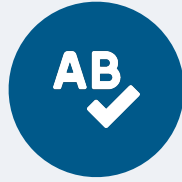
Affordances are offered as **creative opportunities, not required targets**, so evaluators can choose among several realizations of the same bridge.

Four judges, four definitions of a good pun



Comedian

Which candidate is funniest to a native French speaker?



Linguist

Which has the strongest, most recognizable wordplay?



Editor

Which is most natural and fluent in French?



Translator

Which best preserves the humorous intent of the English source?



12 candidates anonymized

5-digit IDs, shuffled, generator metadata stripped



Each persona ranks top 5

rankings, not numeric scores



Weighted Borda count

one winner per generator

Four finalists, three judges, one winner

4 finalists

- claude-sonnet-4.6 (Stage 1 winner)
- gemini-3-flash (Stage 1 winner)
- gpt-5.5 (single candidate)
- gemini-3.1-pro (single candidate)



3 judges × 4 personas

- gpt-5.5
- claude-sonnet-4.6
- gemini-3.1-pro
- *fresh IDs, shuffled, no model metadata*



Aggregation strategies

- Judges then models
- Models then judges
- Pooled: one Borda over all rankings
- Configurable judge and persona weights

We also measure inter-judge agreement, prompt-position effects and self-preference (normal, half-weight and removed-self-vote corrections).

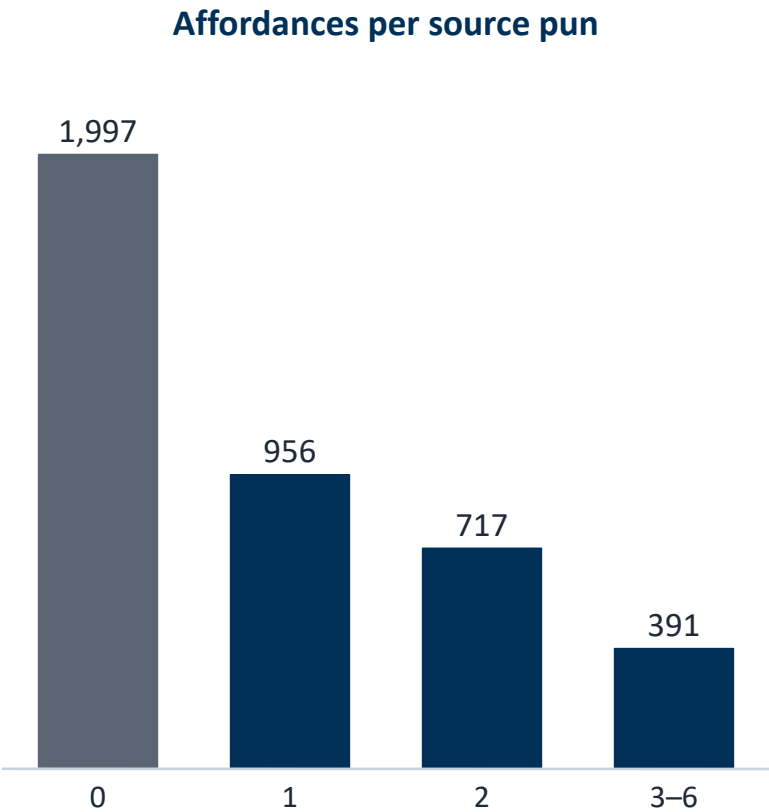
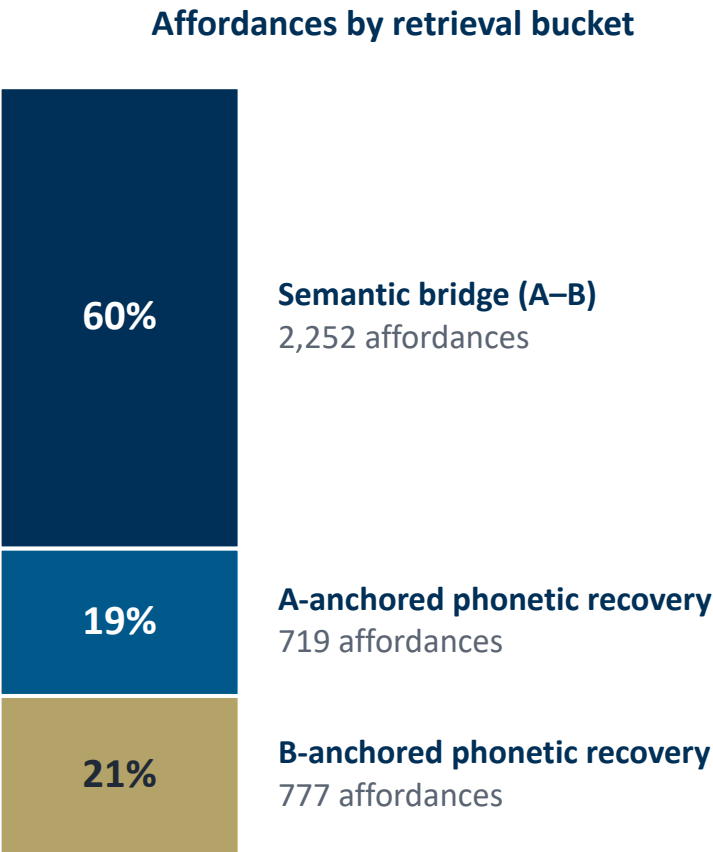
Half of the puns now have a bridge

3,748
affordances retrieved

50.8%
of 4,061 source puns covered

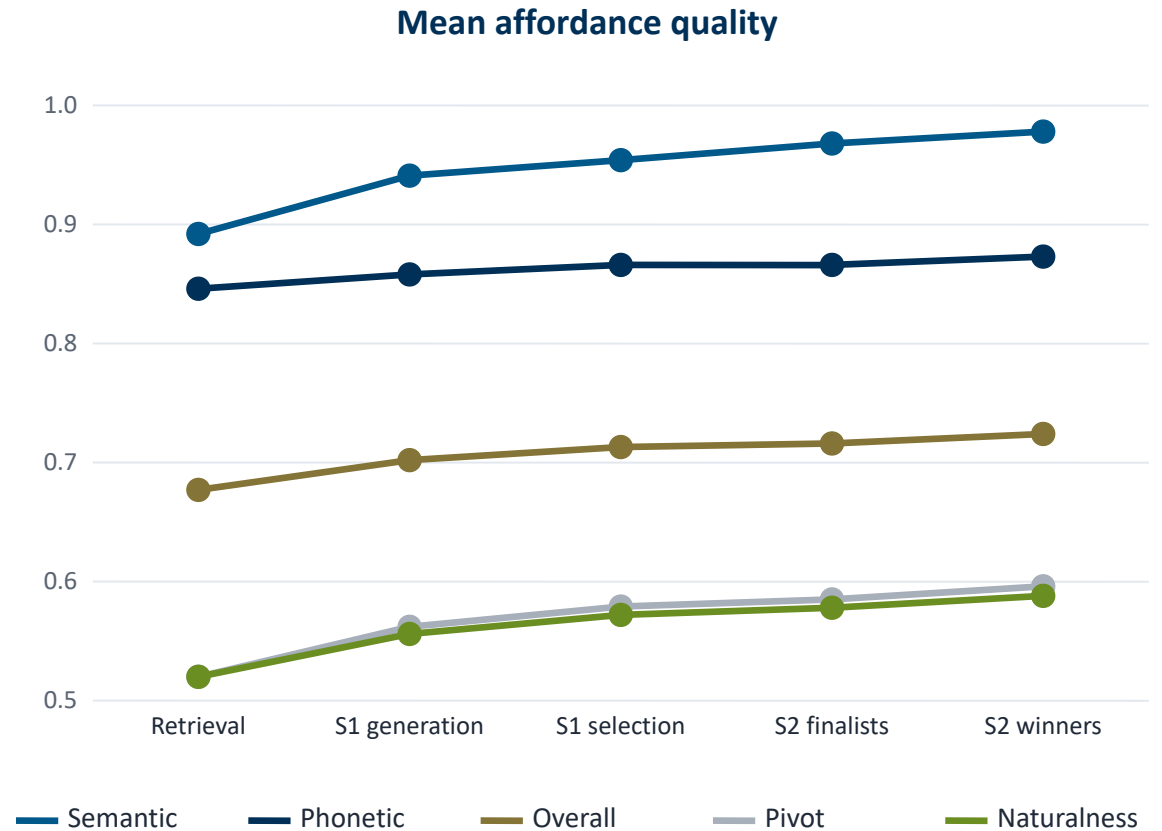
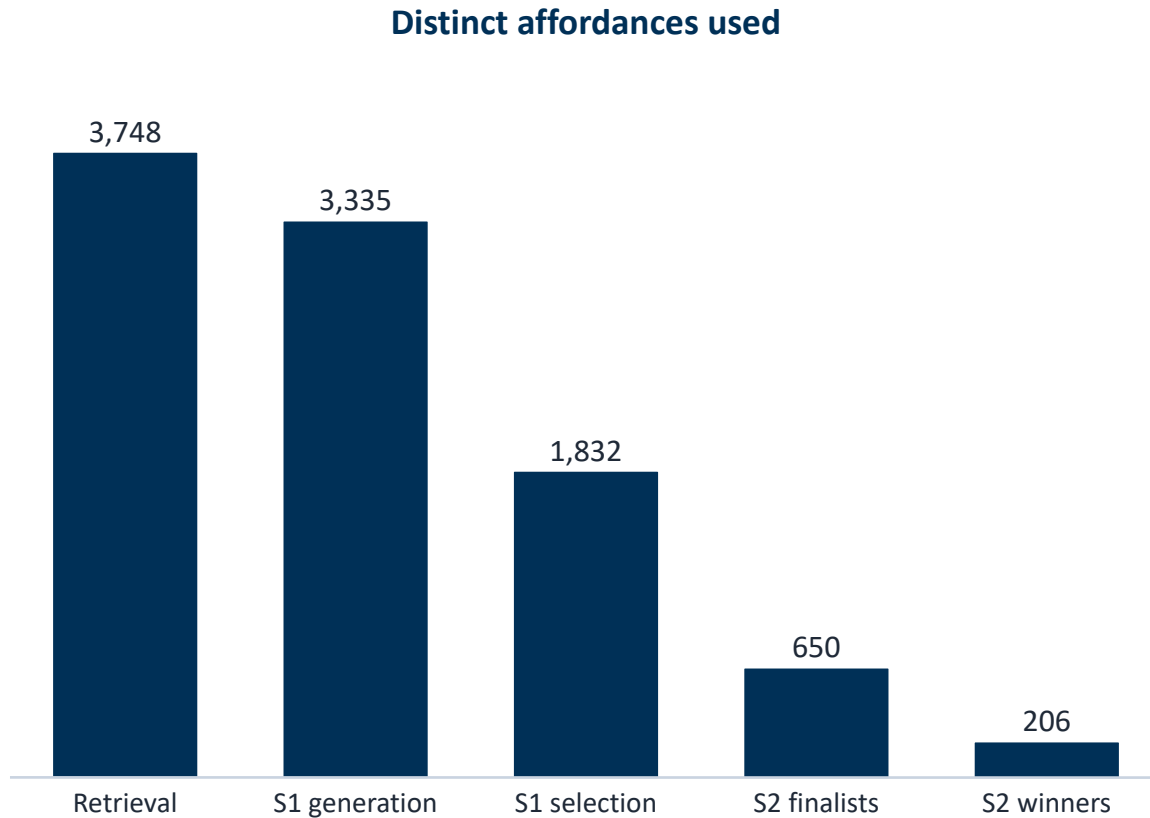
1.82
affordances per covered pun

Homophonic puns 55.1% vs. homographic 50.2% coverage



Same-sound affordances: 10.8% of inventory, but mean score 0.817 vs. 0.660 for similar-sound

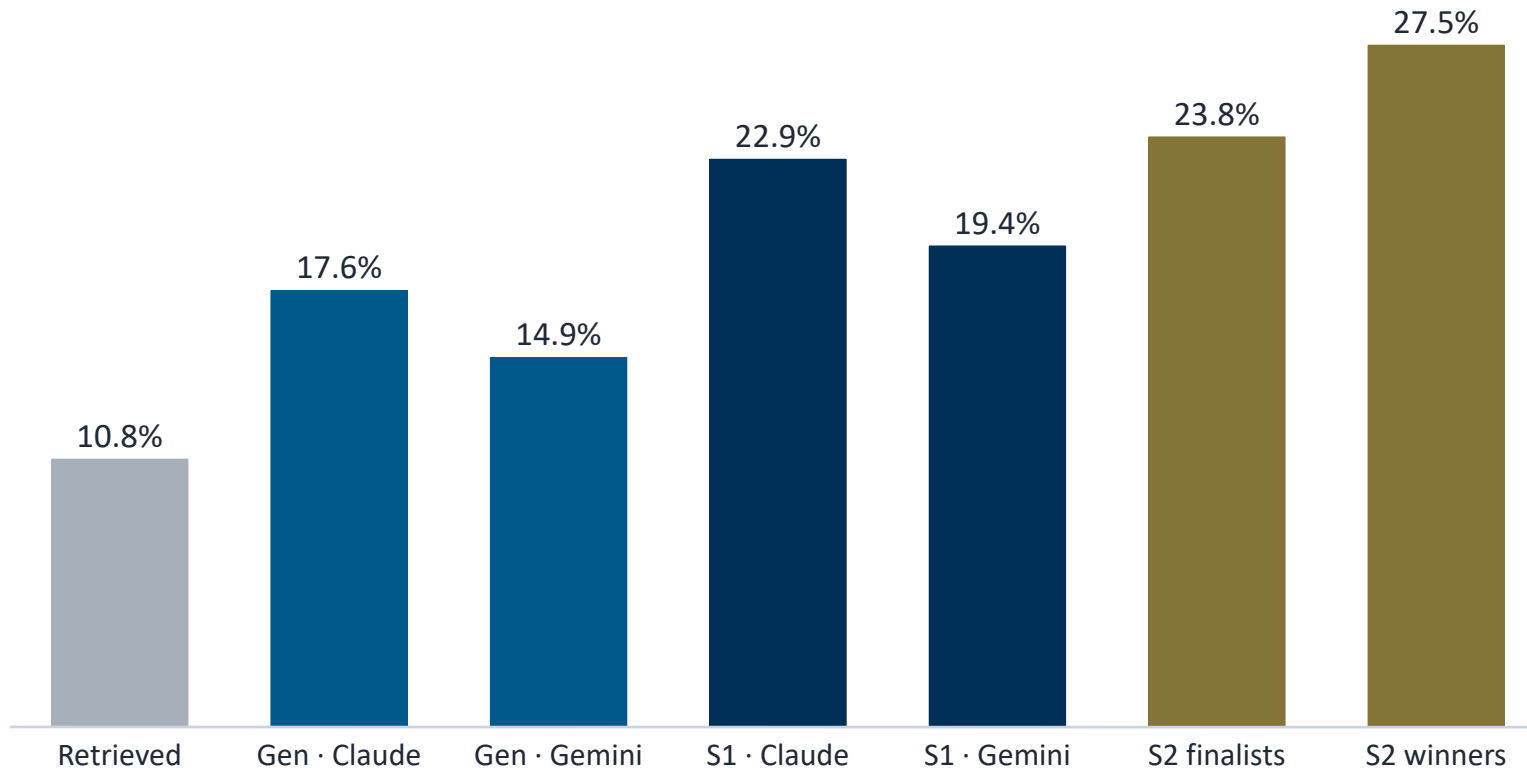
Fewer affordances survive, and the survivors are stronger



Puns using affordances: 2,064 (50.8%) at retrieval → 141 (3.5%) among Stage 2 winners

Exact sound collisions win out of proportion

Same-sound affordances as % of affordance usage



10.8% → 27.5%

share of same-sound affordances, from retrieved inventory to Stage 2 winners

Approximate similarity is useful for exploration; exact collisions are chosen when they exist.

A–B semantic bridges also stay dominant: 60.1% of retrieval, ~72% of finalists and winners.

Judges disagree, and some prefer themselves

16.8%

Claude examples with full
persona agreement in Stage 1

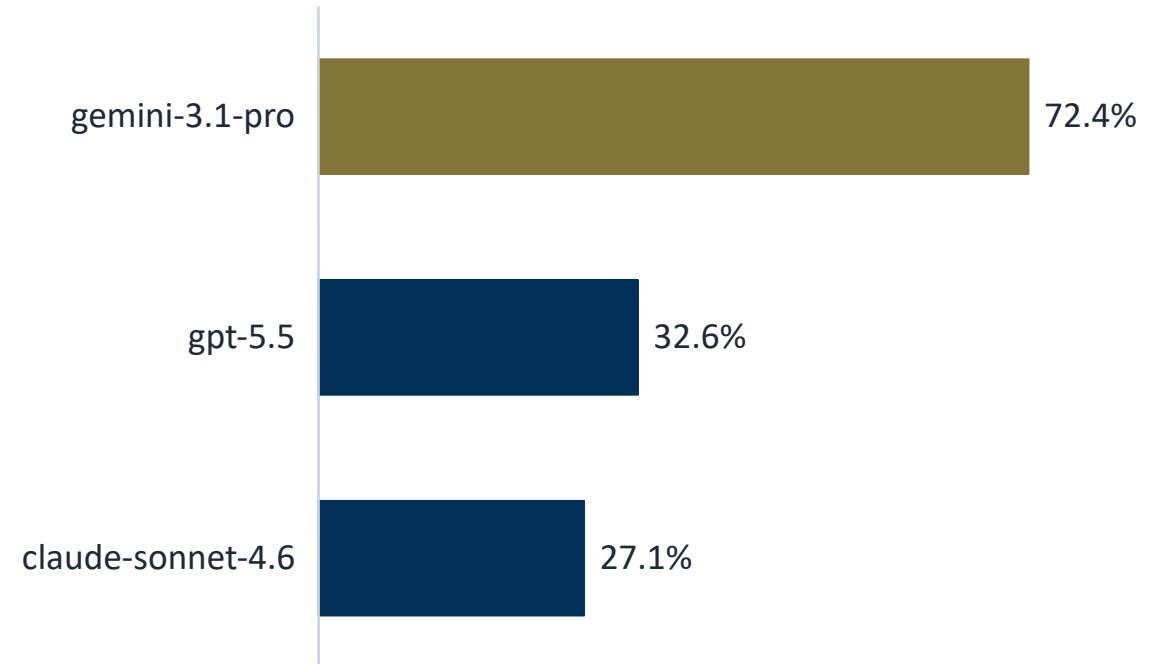
5.4%

Gemini Flash examples with full
persona agreement

0.56–0.86

pairwise Kendall τ between
personas in Stage 2

Judge picks a candidate from its own model family (%)



Gemini family = two of four finalists. Choice of judge and persona weights substantially shifts the final winner distribution.

Results: 1st in human evaluation

44%

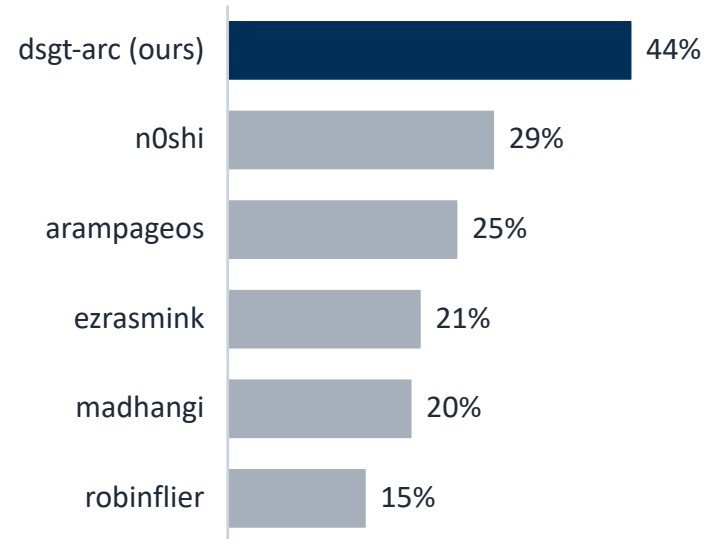
of translations preserved both meaning and wordplay
Ranked 1st, 2nd, and 3rd in manual evaluation

29% best run from the next team

74% our 2025 best, also 1st (different test set and protocol)

Sources: CLEF 2026 JOKER Task 2 overview, Tables 2–4; 2025 overview, Table 7.

Manual evaluation: best run per team



Pun-location metric (official automatic)

Ranks 1–5: 32.59 vs. 28.71 for the next team

Correlation with human success: $\rho = 0.85$

Wordplay vs. meaning (best run)

Team	WP	MP	Success
dsgt-arc	82%	48%	44%
n0shi	38%	70%	29%
arampageos	29%	77%	25%

WP: contains wordplay · MP: preserves the source meaning

BLEU (complementary)

Near the bottom: 21.5 at best

Correlation with human success: $\rho = -0.08$

What we learned



Retrieval matters

Generators use affordances; selection concentrates on the strongest ones.



Still the bottleneck

49% of puns get no affordance; only 3.5% of winners use one.



Same sound wins

Exact collisions are rare but chosen out of proportion.



Few strong bridges

Thousands of affordances, but they reuse a few underlying opportunities.



Position bias persists

Order still influences LLM judges after randomization.



Healthy disagreement

Humor, wordplay, fluency and fidelity pull in different directions.



Wordplay vs. meaning

Most wordplay of any team (82%), least meaning preserved (48%).



Next: search further

Explore more distant semantic and phonological neighborhoods.



Low's approach is still promising

“ Successful pun translation is not the preservation of a word. It is the discovery of another place in the language where sound and meaning collide.

Languages are full of such hidden bridges. The beauty of pun translation lies in finding them.

Low's process, computed

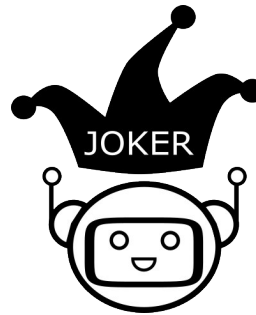
Squares, pentagons, hexagons
→ **semantic expansion**

Sound neighborhoods
→ **learned phonetic space**

Points of contact
→ **affordance bridges**

Translator's judgment
→ **multi-persona ranking**

Thank you!



github.com/dsgt-arc/joker-2026 · {rdtaylorjr, abrikman3, pawate3}@gatech.edu

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